

C A S E S E R I E S

Medical microneedling for the treatment of striae distensae. A prospective study using the exceed Medical Needling Pen

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ABSTRACT

Background and Objectives: Striae distensae, commonly known as stretch marks, are benign linear scars that result from excessive skin stretching, e.g., during puberty, pregnancy, weight loss, or Cushing's syndrome. These scars are difficult to treat due to their atrophic, hypopigmented nature (striae albae) or their erythematous, raised appearance (striae rubrae). Microneedling, a form of percutaneous collagen induction therapy, has shown promise in the treatment of various types of scars. This prospective study evaluates the efficacy of microneedling with the EXCEED Medical Needling Pen™ (MT. DERM GmbH, Germany, Berlin) in the treatment of striae distensae and examines skin tightening, firmness, pigmentation, collagen matrix restructuring, and percutaneous collagen induction.

Materials and Methods: A total of 8 subjects with acute and chronic stretch marks on the body were treated once a month over a treatment period of 60-90 days. Assessments were made at the beginning, before each treatment, and three months after treatment. The documented parameters include gender, age, BMI, Fitzpatrick skin type, and location of striae distensae. Objective skin measurements were performed using the Visioscan® VC 20plus (Courage + Khazaka electronic GmbH, Cologne, Germany) and the Tewa, Mexa, Corneo, Idento, and Cutometer® (Courage + Khazaka electronic GmbH, Cologne, Germany). The measurements compare the skin areas affected by striae with the adjacent healthy skin areas as a control group. Standardized questionnaires (Patient and Observer Scar Assessment Scale "POSAS" and Numeric Rating Scale "NRS") were used by medical staff and the subjects to record subject satisfaction.

Results: The results showed a significant improvement in skin structure, skin smoothness, and wrinkle depth based on the parameters of the anisotropy index, SEsm (sebum amount stratum externum), and SEw (water content stratum externum). The analysis shows that the effect is time-dependent and may vary between patients.



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Based on the results obtained from questionnaires (POSAS, NRS), the medical examiners and subjects reported a subjective improvement in skin texture. For the first time, objective data with a reproducible treatment algorithm for the therapy of striae distensae are presented. Further evidence is provided that microneedling can have a positive effect on skin structure and quality in striae. Due to the small number of subjects, these significant results can be interpreted as trends in treatment effect. Future studies should include a broader participant demographic and additional follow-up intervals to better understand long-term effects and possible recurrence of striae distensae.

Key words: striae distensae, stretch marks, percutaneous collagen induction

Introduction

Striae distensae (SD), colloquially known as stretch marks, are linear dermal scars caused by rapid skin stretching as often occurs during pregnancy, rapid weight gain, or puberty¹. They are divided into two main types: striae rubrae (early, inflamed stage) and striae albae (later, atrophic stage), each of which presents specific therapeutic challenges. The treatment of SD is difficult due to its resistant nature and the limited effectiveness of existing therapies, underscoring the need for new therapeutic approaches. In recent years, microneedling has emerged as a promising, minimally invasive treatment option for improving the appearance of SD. This technique is based on the creation of controlled micro-injuries in the skin which stimulate a wound healing response and neocollagenesis. Alster and Li showed in their long-term study that microneedling can cause a significant reduction in the manifestation of striae distensae in various skin types². Similarly, a pilot study by Park et al. demonstrated the effectiveness of this technique in improving skin elasticity and texture in striae patients³. The results of both studies suggest that microneedling can offer lasting aesthetic improvement for patients. Aust et al. introduced the concept of percutaneous collagen induction therapy as a novel therapeutic approach to stimulate collagen synthesis in the skin. This technique has been described as an alternative to other non-invasive procedures such as lasers and chemical peels and shows significant potential for improving the damaged skin structure in SD⁴. Comparative studies

underscore the effectiveness of microneedling compared to other modalities. Microneedling was compared with fractional CO₂ laser therapy and found that microneedling showed comparable improvement with a lower side effect profile⁵. Furthermore studies found that combining microneedling with platelet-rich plasma (PRP) enhanced treatment outcomes in terms of improving skin structure suggesting a potential for adjuvant therapies to optimize the effects of microneedling⁶. Another focus of current research is the analysis of the biophysical properties of SD and the mechanisms underlying the effects of microneedling. Non-invasive tests to assess the structural characteristics of SD thereby creating a basis for objective evaluation of the treatment effect⁷. Gardien et al. investigated transepidermal water loss (TEWL) in scarred skin and emphasized the importance of the skin barrier function as an indicator of skin integrity in the treatment of SD⁸. Advanced measuring devices such as the Tewameter TM300 and the scientific products from Courage Khazaka make it possible to measure such biophysical parameters precisely and reliably. In addition recent studies shed light on the mechanism of microneedling at the cellular level. Hou et al. described the wound healing cascade triggered by microneedling which stimulates the production of collagen and elastin, thereby promoting the structural integrity of the skin⁹. Microneedling promotes the activation of fibroblasts and the restructuring of the extracellular matrix, which are processes that are particularly relevant for the repair and remodeling of damaged skin in SD. In addition to these physiological effects the psychosocial

burden of SD is also significant as many patients suffer from reduced self-esteem and a reduced quality of life¹⁰. Studies documented the positive psychological effects after SD treatment and found high patient satisfaction after microneedling for the treatment of various dermatological problems including SD^{11,12}.

Materials and Methods

Test subjects

A total of 8 out of 22 adult subjects (7 women and 1 man) with striae distensae (rubae et albae) on the trunk or extremities were included in the prospective study. The subjects ranged in age from 18 to 64 years, with an average age of 38 years and a variation in skin types according to Fitzpatrick (1-4). The inclusion criteria comprised subjects with SD on the body that had existed for less than one year (SD in the acute phase) or more than one year (SD in the chronic phase). Exclusion criteria included skin structure without SD (as microneedling is not necessary for healthy skin), subjects with striae less than six months old and/or previous microneedling therapies for striae in the last six months, skin tumors and precancerous lesions, acute skin diseases (including acne, rosacea, psoriasis, neurodermatitis) and minors. Patients who had demonstrable sun exposure (UV light radiation) in the area being treated were also excluded. There was no exclusion based on the body region where the SD was located.

Treatment with the EXCEED Microneedling Pen and photo documentation

The subjects received up to four therapy sessions (T0, T1, T2, T3) with the EXCEED Medical Needling Pen™ (MT. DERM GmbH, Germany, Berlin) which consists of a tilting plate with six needles (32 gauge) and has a penetration depth of 0.1-2 mm. No additional therapies were included in the analysis. The areas of the body examined and treated with the EXCEED Medical Needling Pen included the hips, thighs and lower abdomen. Before each treatment, lidocaine 23%/tetracaine 7% gel was applied to the treatment area for 30 minutes. The cream was removed

with water-soaked gauze and the skin was degreased and prepared with alcohol immediately before treatment. Aftercare consisted of applying body oil (ENVIRON - Body EssentiA - Vitamin A, C and E Body Oil) after treatment to prevent scabbing. Before the first and after each of the four treatments, as well as during the follow-up examination, the SD was documented with a high-resolution Apple camera and the Visioscan® VC 20plus UV-A video camera (Courage + Khazaka electronic GmbH, Cologne, Germany) as seen in Figure 1 (A-D).

Recording of objective skin parameters

Before each treatment, objective skin parameters were recorded in addition to photo documentation. Objective skin measurements were performed using probes from Courage + Khazaka (Visioscan, Tewameter, Mexameter, Corneometer, Identometer, and Cutometer® (Courage + Khazaka electronic GmbH, Cologne, Germany).

The measurements were taken at the same previously documented location on the treated skin and at control points on the contralateral SD-free skin area as seen in Figure 2 (A). The healthy opposite side represents the desired ideal and provides an original baseline value for the initial examination. To ensure that the measurements of the objective skin parameters were always taken at the same locations (SD and control sites), an individual template adapted to the SD was created for each test subject as seen in Figure 2 (B-C). This was placed on the skin during each therapy session to identify the striae pattern to be treated. The skin area specific to each subject, which served as a control pattern was re-examined and marked before each treatment. The pre- and post-interventional images during the procedure are shown in Figures 9-11.

Visioscan® VC20plus is used to record surface structure, skin smoothness, roughness, flakiness and wrinkles. Important Visioscan parameters are: anisotropy index, SEsm ("sebum amount stratum externum," skin smoothness) and SEw ("water content stratum externum," wrinkles) listed in the below provided Table 1.

The Tewameter measures transepidermal water loss to evaluate the skin barrier function. The

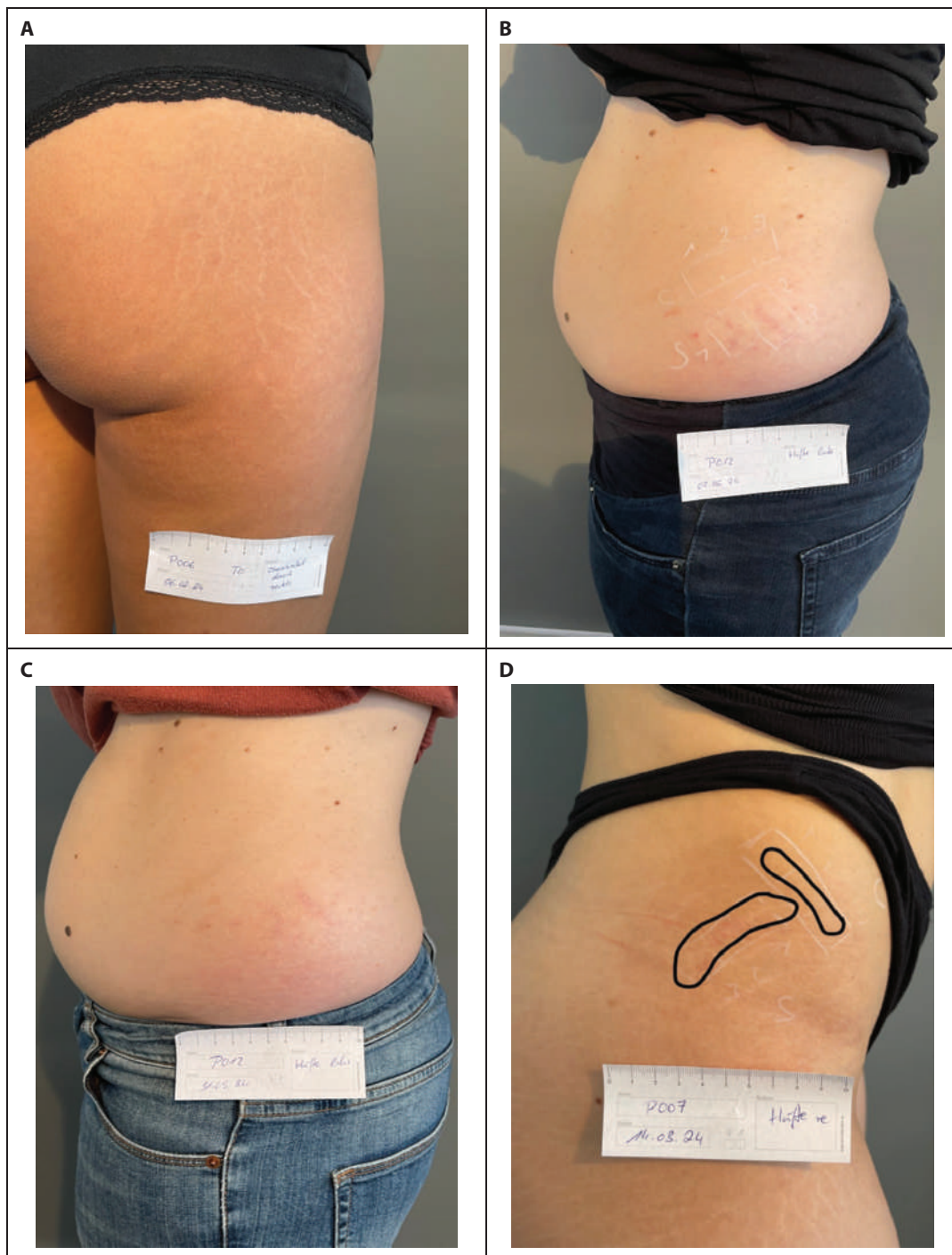


Figure 1. (A-D): (Striae albae et rubae) Standardized photo documentation before each treatment session, before the first microneedling therapy.

Mexameter is used to evaluate the melanin and hemoglobin components. Skin moisture especially in the stratum corneum is measured using a Corneometer. Skin firmness and stiffness are measured using an

Identometer. The Cutometer is used to measure elasticity with different R values that indicate elasticity, firmness, and skin resilience. The Tewa, Corneo, and Cutometer® are validated and established instruments

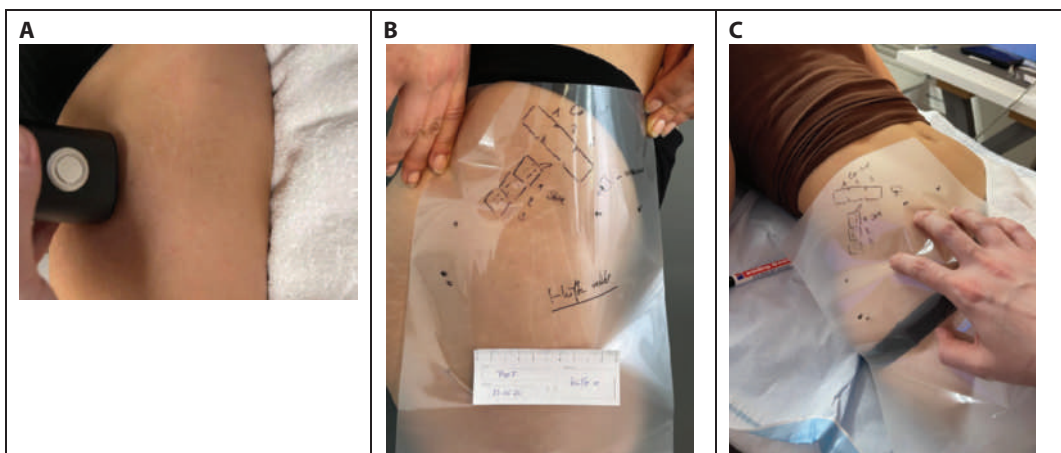


Figure 2. (A-C): (A) Visioscan for objective skin measurement before each microneedling treatment, (B-C) Application of the individual subject template for reproducibility of the SD area to be treated.

for measuring skin structure and in particular skin thickness and elasticity in striae distensae which have already been used^{13,14,15}. The Mexa, Identometer and Visioscan have not yet been used in any scientific studies on the treatment of striae distensae. The parameters can be seen in Table 2. Additional parameters documented in the study include gender, age, BMI, Fitzpatrick skin type and location of striae distensae.

Assessment of quality of life and satisfaction

To assess the quality of life and satisfaction of the test subjects standardized questionnaires (POSAS and NRS) were integrated into the study which were completed before the start of treatment and three months after the last microneedling therapy. The POSAS (Patient and Observer Scar Assessment Scale) is an assessment scale developed to evaluate the quality of scars after an injury or surgery. The POSAS score was introduced by Kirketerp-Møller et al. and has proven useful in clinical practice for assessing both the patient's subjective perception and an observer's assessment of the scar¹⁶. The scale is divided into two parts: the patient assessment and the observer assessment. The patient assessment includes pain, itching, skin color, thickness, and softness of the skin structure. Each of these aspects is rated on a scale from 0 (no discomfort) to 10 (extreme discomfort). The observer assessment in the case of this study by medical staff includes the color, thickness, texture, elasticity and overall impression of

the scar. Here too, the assessment is made on a scale of 0 to 10 with 0 representing an ideal scar and 10 representing a very poor scar¹⁶. The maximum score is 60. The second questionnaire integrated into the study is the Numerical Rating Scale (NRS) which is a widely used tool in medicine for subjective pain measurement¹⁷. The patient is asked to rate their pain on a scale from 0 to 10 where 0 represents no pain and 10 represents the most severe pain imaginable. The NRS is often used to quickly and effectively assess the intensity of pain whether in hospitals, outpatient care or pain therapy. The NRS was first described by Price et al. in 1983.

Statistical analyses

Objective skin parameters

SUBJECT-SPECIFIC EVALUATION

An analysis was performed to determine whether the objectively recorded skin parameters (see section 2.3) differed significantly between the treated striae and the control sites. A total of 64 parameters were evaluated (39 Visioscan parameters and 25 parameters from the other probes). The statistical analyses were performed using the R software package (version 4.4.0, <https://www.r-project.org/>). For each subject and examination time point the relative change in each

Table 1. Visioscan® VC20plus probes used and associated parameters with units (source: own table).

Probe	Parameters	Note
Visioscan® VC20plus	Anisotropy Index	Directional dependence of the lines; the older the skin, the higher the index
	Skin smoothness (SE_{sm})	The smaller the SE_{sm} , the smoother the image
	Skin roughness (SE_r)	The smaller SE_r , the rougher the skin
	Scaliness (SE_{sc})	The smaller SE_{sc} , the less scaling
	Wrinkles (SE_w)	The wider the wrinkles, the higher this value
	Horizontal R1	Roughness depth R1
	Horizontal R2	Maximum roughness depth R2
	Horizontal R3	Average roughness R3
	Horizontal R4	Smoothing depth R4
	Horizontal R5	Arithmetic mean roughness value R5
	Vertical R1	Roughness depth R1
	Vertical R2	Maximum roughness depth R2
	Vertical.R3	Average roughness R3
	Vertical.R4	Smoothing depth R4
	Vertical.R5	Arithmetic mean roughness value R5
	Circular R1	Roughness depth R1
	Circular R2	Maximum roughness R2
	Circular R3	Average roughness R3
	Circular R4	Smoothing depth R4
	Circular R5	Arithmetic mean roughness value R5
	Avg, H/V R1	
	Avg, H/V R2	
	Avg, H/V R3	
	Avg, H/V R4	
	Avg, H/V R5	
	Contrast	Texture parameters
	Energy	Texture parameters
	Entropy	Texture parameters
	Homogeneity	Texture parameters
	Variance	Texture parameters
	Volume	
	Surface	
	All Classes Cell Count	Young skin shows more cells than aged skin
	Cell Count Class 1	0.01-0.05 mm ²
	Cell Count Class 2	0.05-0.1 mm ²
	Cell Count Class 3	0.1-0.2 mm ²
	Cell Count Class 4	0.2-0.5 mm ²
	Cell Count Class 5	0.5-1 mm ²
	Cell Count Class 6	≥ 1 mm ²

Table 2. Probes used by Courage + Khazaka and associated parameters with units (source: own table).

Probe	Parameter	Unit
Corneometer® CM 825 (Electronic capacity between two electrodes)	Moisture / Hydration Stratum corneum (10-20µm)	%
Indentometer IDM 800 (Resistance of the skin to mechanical compression)	Skin firmness/stiffness / Indentometer value	mm (penetration depth of sensor/ stamp into the skin in mm)
Tewameter® TM 300 (Water loss rate determined by measuring water vapor density at 2 heights above the skin)	Transepidermal water loss (TEWL)	g/m²/h (grams of water loss per m ² of skin per hour)
Mexameter® MX 18 (spectrophotometry/ Remission measurement)	Melanin	relative unit (0-999)
Mexameter® MX 18 (spectrophotometry/ Reflection measurement)	Hemoglobin / Erythema	Relative unit (0-999)
Cutometer® MPA 580 (Elastic skin property through negative pressure (skin is sucked in and the deflection (skin deformation in mm) over time (R1-R8) is recorded for elasticity, viscoelasticity, and relaxation, among other things)	Rx: Firmness (R0) (skin deformation at time R1-R8)	mm (Skin deformation in mm through negative pressure measurement)
	Rx: Residual deformation (R1 and R4)	mm
	Rx: Viscoelasticity (R2)	%
	Rx: Fatigue effects (R3, R9, and R10)	mm
	Rx: Net elasticity (R5)	%
	Rx: Proportion of viscoelasticity (R6)	%
	Rx: immediate recovery (R7)	mm
	Rx: Total regression (R8)	mm
	Fx: Skin elasticity (F0/F1)	mm*s (Skin deformation in mm per second measured by negative pressure)
	Fx: Skin deformation during the second suction cycle in multi-cycle measurement (F2)	mm*s
	Fx: Skin energy (F3)	mm*s
	Fx: Firmness (F4)	mm*s
	Qx: Maximum relaxation area (Q0)	mm*s
	Qx: Total relaxation fraction (Q1)	%
	Qx: Elastic relaxation fraction (Q2)	%
	Qx: Viscoelastic relaxation component (Q3)	%

skin parameter was calculated based on three control sites and three treated striae distensae (SD). For each SD the relative changes to all three control sites were determined. The relative change expresses the percentage by which the measured value for the treated striae

deviates from the control value (e.g., -20% corresponds to a value that is 20% lower than that of intact skin). This results in nine values per subject and examination time point (3 SD × 3 controls). T0 represents the time point before the first treatment (day 0), T1 after the

first and before the second treatment, T2 is the time point after the second treatment and before the third treatment and T3 reflects the time point after the third treatment. With 2, 3 and 4 examination time points there are 18, 27 and 36 measured values per subject respectively. For each subject a linear regression analysis was performed based on the relative changes to model the relationship between days after initial treatment and the change in the respective skin parameter. The significance of a change over the observation period was assessed using the slope parameter (p-value) and a p-value ≤ 0.01 was defined as significant. In addition the estimated relative change in skin parameters at 0, 30, 60 and 90 days and the corresponding 95% confidence interval were calculated from the regression model. The evaluation was performed separately for each subject.

CROSS-SUBJECT EVALUATION

Based on the relative changes determined for each subject additional cross-subject analyses were performed. For the time points 0, 30, 60 and 90 days after initial treatment the mean relative change for each parameter and the corresponding 95% confidence interval were calculated based on the regression analysis. The cross-subject effect was estimated using the MU/Hampel method taking into account the uncertainty of the regression¹⁸.

POSAS SCORE

The statistical analyses were performed using SPSS.

Results

Subjects and treatment times

Of the 22 subjects 8 could be included in the study due to loss to follow-up. Here 4 subjects showed striae on the lower abdomen and 4 subjects on the lower extremities at the hip or thigh. The change in striae was measured using Visioscan and the five probes mentioned above. Visioscan parameters were recorded for time points (T0-T3) with 1 subject up to T1, 4 subjects

up to T2 and 3 subjects up to T3. For 7 of the 8 subjects parameters were also recorded at each examination time point using Cuto, Corneo, Mexa, Indento and Tewameter. For 1 subject, these values were only available for the initial examination at time point T0. For this subject only the Visioscan parameters were evaluated seen in Table 3.

Significant changes in objective skin parameters are highly dependent on the individual test subjects

Figure 3 provides an overview of the number of skin parameters for which a significant increase or decrease was observed during the treatment period. The results are presented separately for the Visioscan and Cutometer probes and aggregated across all probes used. Overall it can be seen that the number of significant changes varies greatly between individual subjects. While for example a significant increase or decrease was observed in 75% of the skin parameters examined over time in subject P003 significant changes were only detectable in 27% of the parameters for other subjects, such as P009. These marked differences underscore that the response of objective skin parameters to treatment is highly individual and subject to considerable interindividual variation.

In the individual analysis of the subjects, trends were already apparent for the anisotropy index, SEsm and SEr. Here, among other things the anisotropy index showed a decrease over the treatment period of 60–90 days after microneedling treatment, as shown in the Figures 4 and 5 below for P008.

Some skin parameters show a significant effect across all test subjects

For the Visioscan parameters anisotropy index, skin smoothness (SEsm), skin tightness (SEr), flakiness (SEsc) and wrinkles (SEw) the results of the cross-subject analysis for time points 0, 30, 60 and 90 days after the initial treatment are summarized in Table 4. In addition the individual values (blue bars) with a robust overall mean (blue line) and 95% confidence interval (light blue band) are shown in Figures 6–8. For the anisotropy index it can be observed that the

Table 3. Overview of the test subjects examined, including the area of the body treated, the date and time of the examination, and the measuring instruments used. Of a total of 22 test subjects, 8 were included in the study.

Patient	Position	Date	Time (days after T0)	Visioscan	Cuto-, Corneo-, Indento- Mexa-, and Tewameter
P003	Right hip	February 20, 2024	T	✓	✓
		03/27/2024	T1 (36 days)	✓	✓
P006	OS right	02/06/2024	T	✓	✓
		03/05/2024	T1 (28 days)	✓	✓
		Apr. 16, 2024	T2 (70 days)	✓	✓
P007	Right hip	02/05/2024	T	✓	✓
		03/14/2024	T1 (38 d)	✓	✓
		May 23, 2024	T2 (108 d)	✓	✓
P008	Lower abdomen, right side	02/26/2024	T	✓	✓
		03/25/2024	T1 (28 days)	✓	✓
		Apr 23, 2024	T2 (57 days)	✓	✓
		May 27, 2024	T3 (91 days)	✓	✓
P009	Lower left abdomen	02/27/2024	T	✓	✓
		March 26, 2024	T1 (28 days)	✓	✓
		Apr. 29, 2024	T2 (62 days)	✓	✓
		May 27, 2024	T3 (90 days)	✓	✓
P010	Lower abdomen, right side	02/28/2024	T	✓	✓
		03/26/2024	T1 (27 days)	✓	✓
		May 8, 2024	T2 (70 days)	✓	✓
P011	Lower abdomen	02/29/2024	T	✓	✓
		March 25, 2024	T1 (25 days)	✓	✓
		Apr. 23, 2024	T2 (54 days)	✓	✓
		May 21, 2024	T3 (82 days)	✓	✓
P012	Left hip	07.05.2024	T	✓	✓
		May 31, 2024	T1 (24 d)	✓	✗
		June 25, 2024	T2 (49 days)	✓	✗

cross-patient mean decreases over the period considered, i.e. a decrease in striae can be observed over the course of the treatments and thus an approximation to the value of the control sites. A similar observation can be made for the parameters SEsm (skin smoothness) and SEw (wrinkles). The anisotropy index is particularly noteworthy. From day 60 onwards, a significant improvement in the parameters is observed, reflecting in an improvement in striae under treatment with the EXCEED Medical Needling Pen. The anisotropy index is a parameter that describes the directional

dependence of the skin's line structure. The older the skin, the higher the anisotropy index. The index indicates how pronounced the lines of the skin are in a certain direction. A higher value indicates a stronger directional dependence which is typically the case with older or damaged skin¹⁹. A decrease in the value during the course of treatment therefore indicates that the directional dependence is decreasing. The SEsm feature refers to skin smoothness and is the value of skin oil (sebum) and measures the surface texture of the skin. The parameter is often used in dermatology and

Value	P003	P006	P007	P008	P009	P010	P011	P012
All probes								
# Parameter ↗	18	18	5	12	3	13	9	15
# Parameter ↘	30	6	23	23	14	13	35	0
# Parameter total	64	64	64	64	64	64	64	39
% Parameter ↗	28%	28%	8%	19%	5%	20%	14%	38%
% Parameter ↘	47%	9%	36%	36%	22%	20%	55%	0%
% Parameter ↗ ↘	75%	38%	44%	55%	27%	41%	69%	38%
Cutometer® MPA 580								
# Parameter ↗	10	6	0	7	0	5	0	
# Parameter ↘	3	3	1	2	10	5	9	
# Parameter total	20	20	20	20	20	20	20	
% Parameter ↗	50%	30%	0%	35%	0%	25%	0%	
% Parameter ↘	15%	15%	5%	10%	50%	25%	45%	
% Parameter ↗ ↘	65%	45%	5%	45%	50%	50%	45%	
Visioscan® VC20plus								
# Parameter ↗	6	11	4	5	1	6	6	15
# Parameter ↘	26	2	22	20	4	7	26	0
# Parameter total	39	39	39	39	39	39	39	39
% Parameter ↗	15%	28%	10%	13%	3%	15%	15%	38%
% Parameter ↘	67%	5%	56%	51%	10%	18%	67%	0%
% Parameter ↗ ↘	82%	33%	67%	64%	13%	33%	82%	38%

Figure 3. Overview of the number of parameters (previously shown in Table 2) with a significant increase (red arrow) or decrease (yellow arrow) separately for each test subject. Absolute numbers (#) and proportions as relative frequency (%) of parameter changes. The parameters correspond with each other.

cosmetic research to quantify the texture and appearance of the skin. SEsm is usually read by special measuring devices that analyze the micro-relief structure of the skin. Smooth skin is often less rough and less irregular and correlates with skin health and a younger appearance^{20,21}. The decrease observed in this study therefore suggests that the treatment makes the skin smoother. The SEw (Wrinkles) parameter refers to the measurement of wrinkles and is an important indicator of skin aging and texture. The SEw value quantifies the depth, density and appearance of wrinkles on the skin surface. The lower the SEw result, the lower the number and intensity of wrinkles^{20,22}. The decrease in value in this study indicates that the number and intensity of wrinkles is decreasing. Some parameters such as skin roughness (SEr) and flakiness (SEsc) show no consistent significant change over time. These measurement parameters are, among other things, validated analysis

values for assessing SD. Kim et al. for example used the anisotropy index to measure the viscoelastic behavior of SD similar to Khatyr et al.^{23, 24}. These studies do not directly mention SEsm or SEw in relation to SD. These parameters originate from the SELS/Visioscan system which is typically used in anti-aging studies^{25,26}. Currently there are no known studies that explicitly use SEsm or SEw as measurement parameters for striae distensae. In summary treatment shows a significant improvement in skin quality in striae for certain parameters (especially the anisotropy index). The effect is time-dependent and varies between patients. Overall, the evaluation suggests that microneedling therapy with the EXCEED Medical Needling Pen can have a positive effect on skin structure and quality in striae. Due to the small number of subjects these significant results can be interpreted as trends in treatment effect.

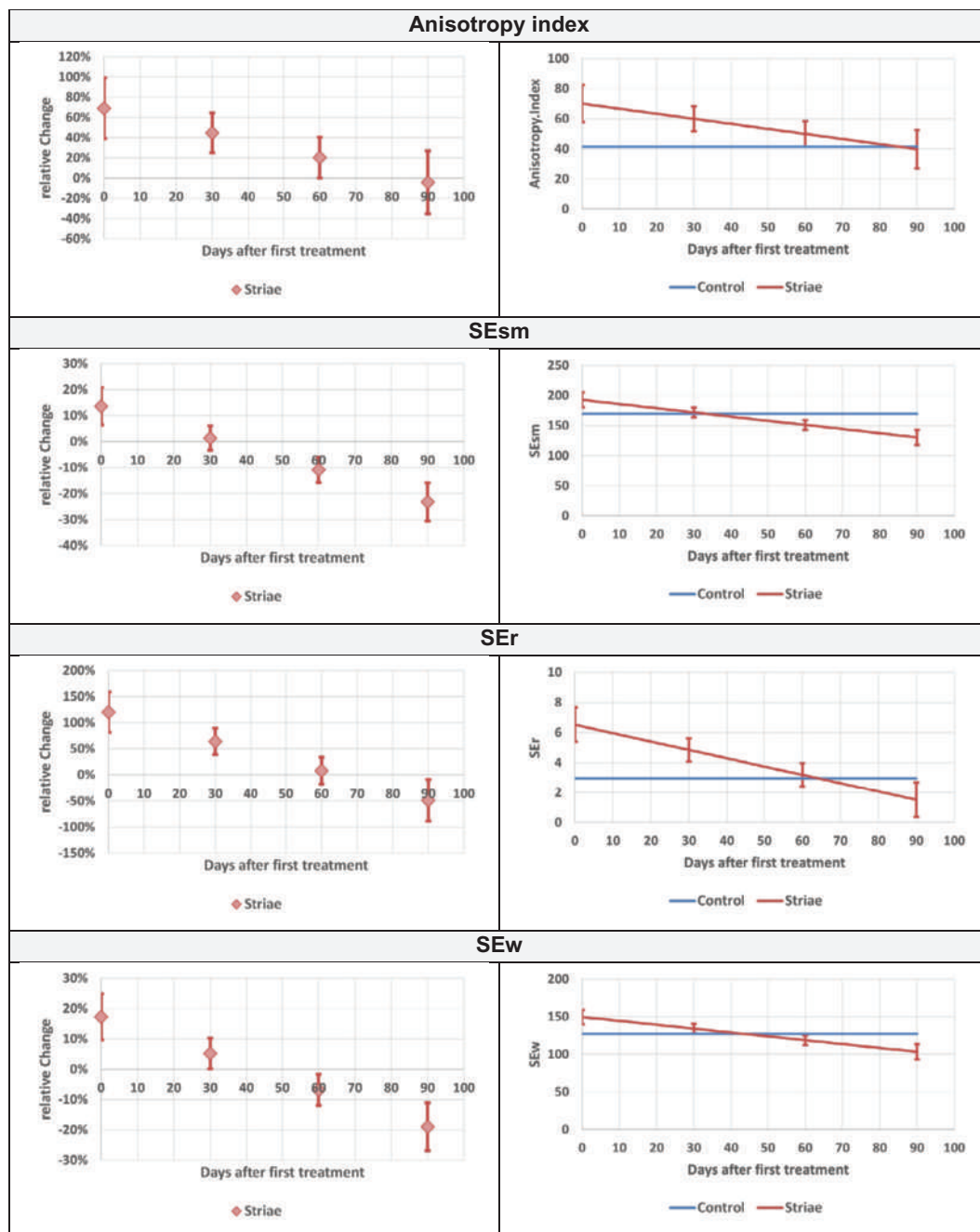


Figure 4. Absolute and relative change for the corresponding skin parameter for test subject P008. The baseline at time T0 represents absolute values for control sites and, in comparison, the corresponding absolute values for stretch marks. Error bars represent the 95% confidence interval.

POSAS score

A statistical evaluation of the POSAS score was performed in the cohort of 8 subjects examined

as seen in Table 5. The maximum score for 6 items was 60 points. A statistically significant improvement in scar quality was observed over the treatment period. The POSAS Observer Score decreased on

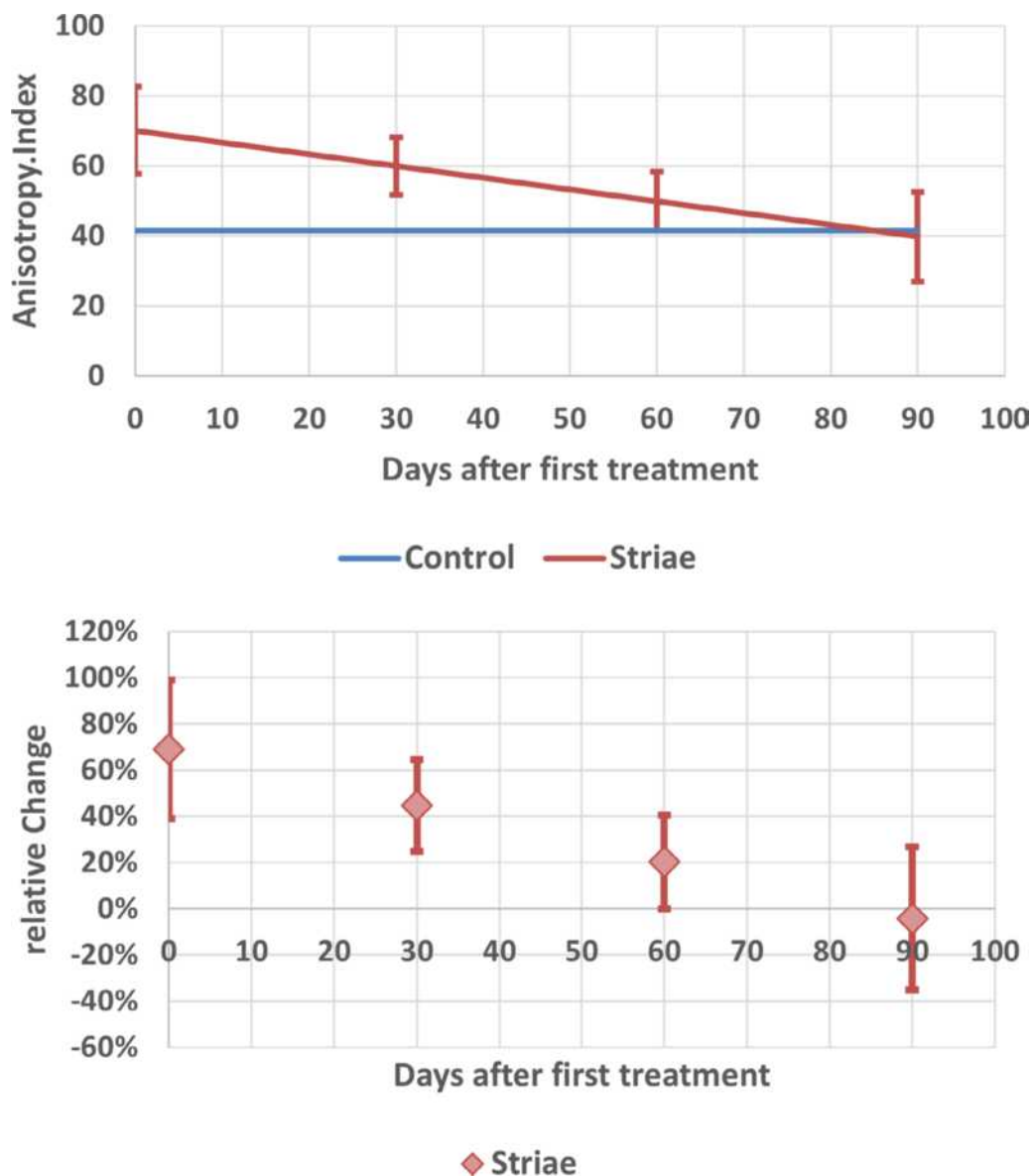


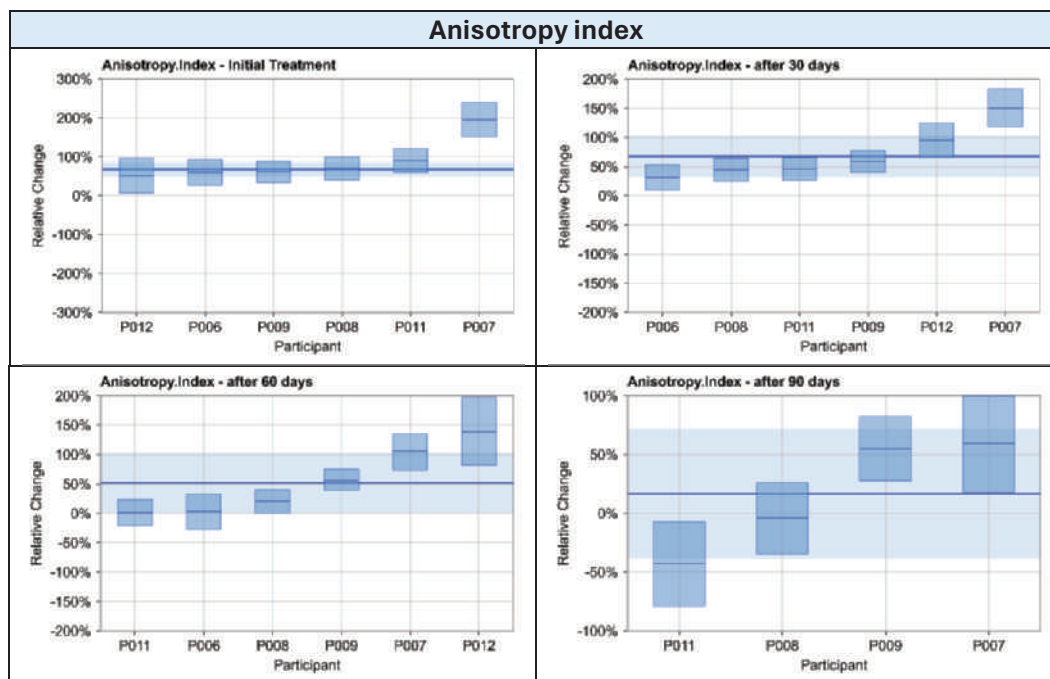
Figure 5. Absolute and relative change for the decreasing anisotropy index for patient P008, for example. The baseline at time T0 represents absolute values for control sites and, in comparison, the corresponding absolute values for stretch marks (upper section of the diagram).

average from 39.50 ± 9.85 to 24.67 ± 5.79 ($p < 0.05$, paired t-test). Similarly, the POSAS Patient Score showed a significant improvement (32.00 ± 7.50 to 22.67 ± 5.79 , $p < 0.05$). The greatest improvements were reported in the subcategories of pigmentation, relief and overall impression. The following results were obtained for the POSAS Observer Score: Before treatment (X) 25-35, striae are noticeable

but not pathological. Depending on depth/color/dermatotype values around 30 ± 5 are common. After treatment (Y): 15-22 a visible but not yet complete improvement. Common values are $18-21 \pm 4$ depending on the treatment. An example here is $30.1 \pm 4.2 \rightarrow 18.3 \pm 3.9$ ($p < 0.05$). There was no change in the items "elevation". Before treatment (X) the POSAS patient score was often slightly lower than

Table 4. Cross-subject mean of the relative change (%) for selected parameters compared to the control site of the subjects.

Probe	Parameter	day	Robust Average	
			Relative Change (%)	95% CI
Visioscan® VC20plus	Anisotropy Index	0	71%	[39%,104%]
		30	50%	[18%,83%]
		60	43%	[-4%,91%]
		90	17%	[-39%,72%]
	Skin smoothness (SE_{sm})	0	44%	[8%,79%]
		30	28%	[2%,54%]
		60	22%	[-4%,48%]
		90	22%	[-44%,88%]
	Skin roughness (SE_r)	0	62%	[-6%,130%]
		30	40%	[-14%,94%]
		60	8%	[-22%,39%]
		90	37%	[-83%,158%]
	Scaliness (SE_{sc})	0	108%	[-85%,302%]
		30	184%	[-7%,376%]
		60	242%	[-8%,493%]
		90	298%	[-188%,783%]
	Wrinkles (SE_w)	0	52%	[16%,88%]
		30	46%	[18%,74%]
		60	44%	[12%,77%]
		90	30%	[-24%,84%]

**Figure 6.** Cross-subject relative change in the anisotropy index for the time points of the initial examination on day 0 and on days 30, 60, and 90 after Microneedling therapy in comparison to the control side (anisotropy index).

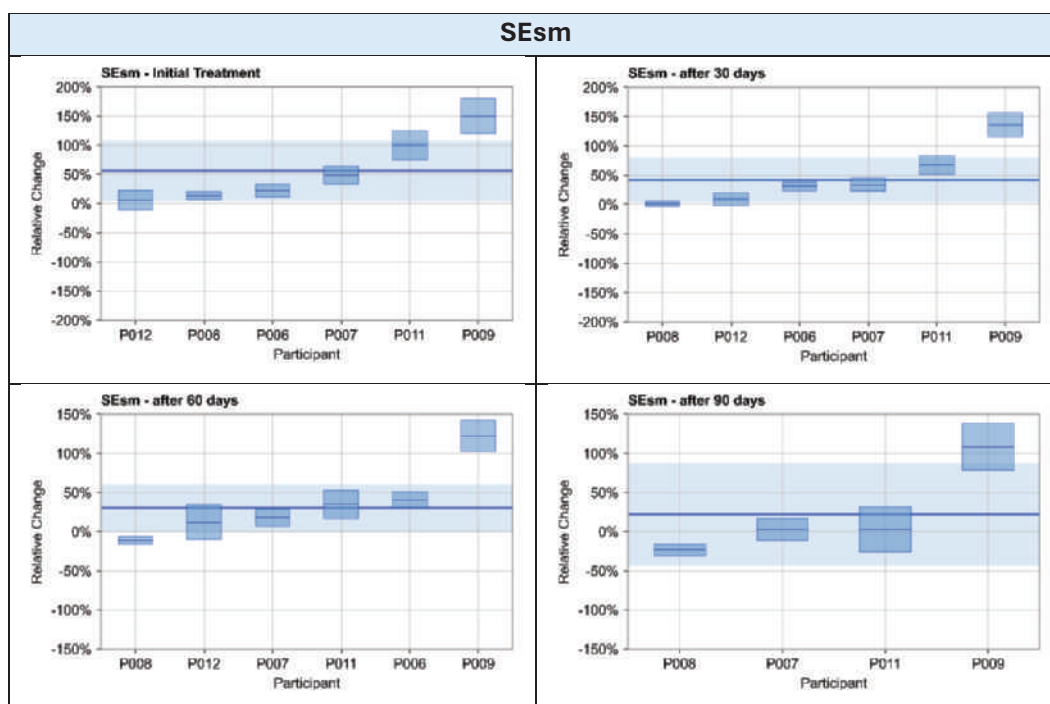


Figure 7. Relative change in SEsm across all test subjects for the initial examination on day 0 and on days 30, 60, and 90 after Mireoneedling therapy compared to the control side (SEsm).

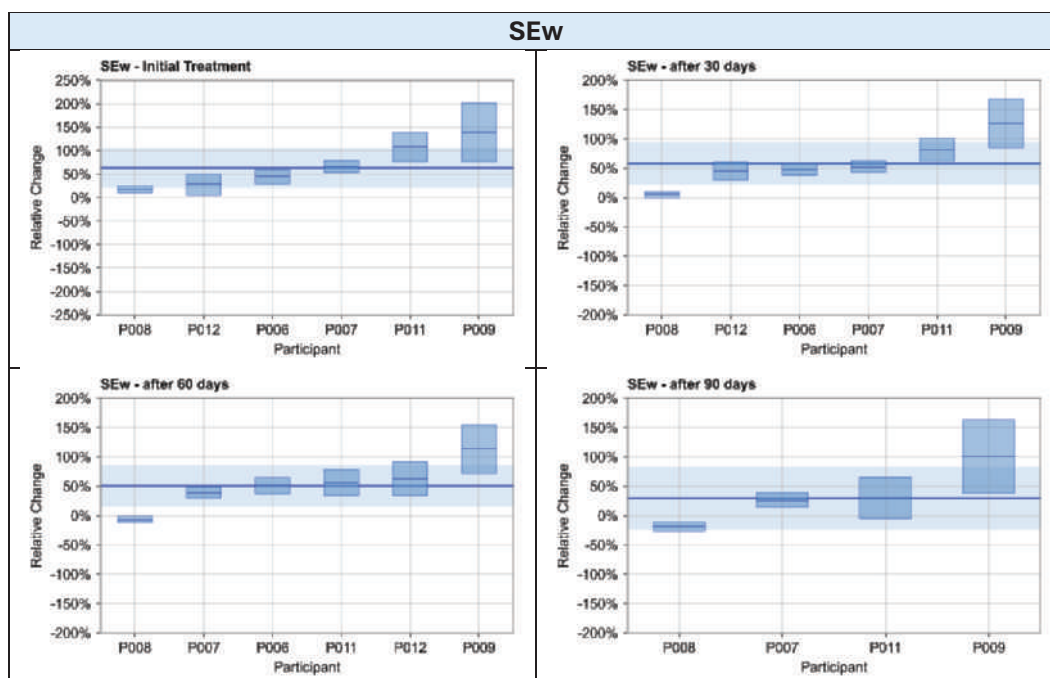


Figure 8. Relative change in SEw across all subjects for the initial examination on day 0 and on days 30, 60, and 90 after microneedling therapy compared to the control side (SEw).

Table 5. POSAS assessments for the respective treatment period (T0-T3). Rated on a scale of 1 to 10 (1 = normal skin, 10 = worst scar imaginable) Total score from 6 to 60 (observer) or 6 to 60 (patient). Overall assessment (last column) is not included in the total.

Subject Nr.	POSAS Patient	POSAS Observer	Assessment
P001	T0 18/4	T0 29/5	
P003	T0 26/7 T1 37/6	T0 34/6 T1 34/6	T1 2
P006	T0 40/10 T1 23/4 T2 18/4	T0 58/9 T1 27/4 T2 24/4	T1 3 T2 3
P007	T0 21/6 T1 20/4 T2 17/4	T0 28/4 T1 25/4 T2 25/4	T1 3 T2 3
P008	T0 33/7 T1 22/5 T2 20/3 T3 15/3	T0 41/7 T1 37/6 T2 33/5 T3 26/4	T1 3 T2 3 T3 3
P009	T0 33/9 T1 37/5 T2 34/4 T3 24/4	T0 48/8 T1 34/4 T2 30/3 T3 17/2	T1 3 T2 3 T3 3
P010	T0 23/5 T1 29/7 T2 28/5	T0 26/5 T1 38/7 T2 20/4	T1 4 T2 3
P011	T0 43/10 T1 42/9 T2 39/8 T3 29/8	T0 43/9 T1 41/9 T2 37/8 T3 31/7	T1 2 T2 2 T3 2
P012	T0 37/7 T1 23/2	T0 38/7 T1 22/4	T1 3

the observer score. Typical values are 22-30, e.g., 28.0 ± 6.1 . After treatment (Y) the patient score was 15-22, often lower than the observer score.

Discussion

Microneedling induces neocollagenesis and remodeling of the extracellular matrix through mechanical stimulation of the dermis which is consistent with earlier studies on percutaneous collagen induction⁵. The documented increase in collagen I and elastin reflects findings by Hou et al. who described the activation of fibroblasts by microtrauma¹⁰. Compared to fractional CO2 laser therapy microneedling shows a comparable efficacy profile with a lower rate of side effects⁶. The combination of Visioscan and various other measuring probes, such as Cutometer measurements offers for the

first time an innovative multimodal evaluation system for SD that supplements subjective scales (including POSAS) with objective biophysical parameters. This expands on the analysis methods and enables precise quantification of the success of therapy⁸. The POSAS score is used here in an exploratory capacity, as formal psychometric validation (e.g., Cronbach's α , ICC) was not performed within the current study^{17,26,27,28}. In future studies the BODY-Q Stretch Marks Scale can be added for extended validation²⁹. In addition the use of individual templates for the test subjects and markings prior to microneedling treatments provides reproducible results and the initiation of a standardized procedure which has been lacking in the literature to date. The high patient satisfaction rate (87.5%) underscores the psychosocial benefits of the treatment confirming earlier findings by SD patients. Although the small sample size (n=8) must be taken into account



Figure 9. (A) Photo documentation before 1st MN treatment, (B) Photo documentation before 2nd MN treatment, (C) Photo documentation after 2nd MN treatment, (D) Measurement points.

as a significant limiting factor here. However the lack of pigmentation change indicates a selective effect of the procedure which primarily affects the dermal structure not epidermal melanocyte activity^{1,11,29}. The prospective study analyzed stretch marks on the lower extremities so future research should include stretch marks on the upper and lower extremities in order to evaluate further information about the effectiveness of microneedling on fine and wider stretch marks. Areas that are frequently exposed to the sun, such as the face

or hands, should be included in future studies. The current study suggests that microneedling is effective for striae distensae regardless of the body region³⁰. Differences in skin thickness or structure do not appear to significantly influence treatment results². Many studies mention needle lengths of 1.5 mm (dermal roller, dermapen) for the treatment of SD with significant improvement in the clinical and histopathological results of the dermis^{3,31,32}. Comparative studies are available for 2.5 mm depth in the area of post-acne scars³³. In this

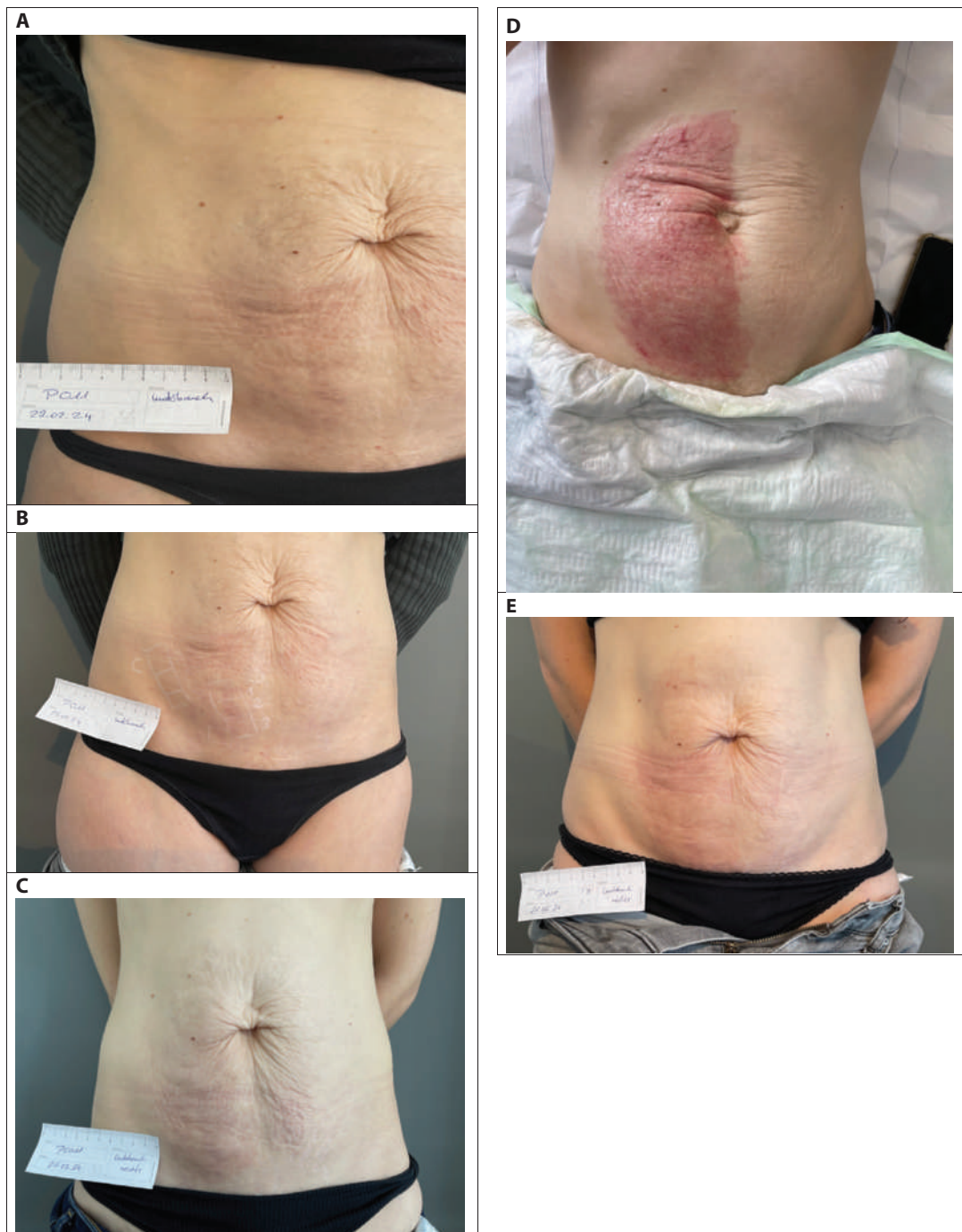


Figure 10. (A-E): (A) Photo documentation before 1st MN treatment, (B) Photo documentation before 1st MN treatment with SD marking, (C) Photo documentation before the second MN treatment, (D) Photo documentation after 2nd MN treatment, (E) Photo documentation after completion of treatment (3x MD therapy).

study 2.5 mm needles were more effective than 1.5 mm needles in the treatment of deep acne scars. However further studies are needed to determine the direct application to striae distensae. Further comparative

studies are needed to identify subtle regional differences^{2,34}. Despite the promising results the most significant limitation is the small sample size (n=8) and the lack of an additional control group which limits

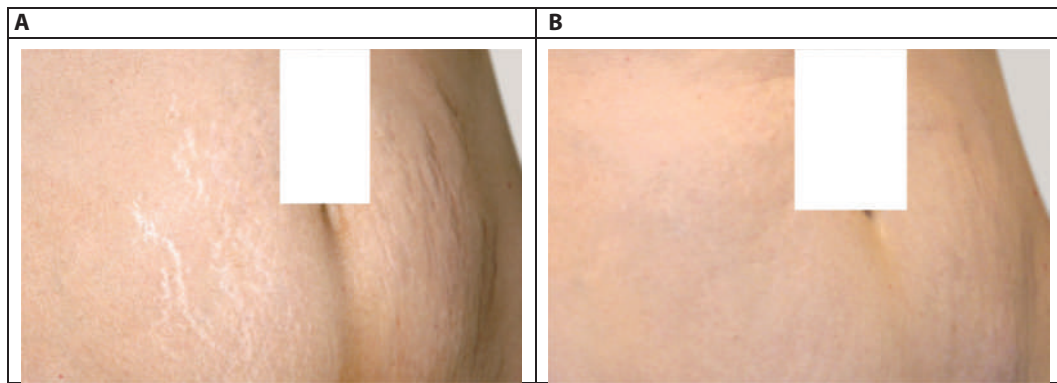


Figure 11. (A) Photo documentation before 1st MN treatment. Anonymized by censor bars, (B) Photo documentation after MN treatment. Anonymized by censor bars.

the generalizability of the findings. Due to loss to follow-up, loss to follow-up bias or attrition bias cannot be completely ruled out. Further studies should integrate bigger test groups with a similar demographic background and an inter- or intraindividual control group to improve the causal interpretation. Measuring points in the control group should be non-treated pre-existing striae. With regard to the patient-specific analysis the number of significant parameters varies between subjects and no generalized picture can be presented due to demographic variability like age, Fitzpatrick skin type, acute or chronic phase of striae. Recent reviews discussed this variability relating to the results after medical needling in striae^{21,35}. In subject P003 in particular approximately three-quarters of the parameters (75%) show a significant change. In contrast, only 27% of the parameters can be observed to change in subject P009. It should also be noted that in six of the eight patients the proportion of parameters showing a decrease over time is significantly greater than the proportion of parameters showing an increase. Future studies should implement randomized designs with larger cohorts and long-term follow-up to test the sustainability of the effects. A significant decrease over time was observed for the anisotropy index, skin smoothness (SEsm) and wrinkles (SEw), indicating that the striae values are approaching the values of the control sites. This suggests that microneedling treatment improves skin structure. A decrease in the anisotropy index during the course of microneedling treatment suggests that the skin structure improves and approaches the values of intact skin. The

anisotropy index in particular is the most important parameter in this study as the hypothesis regarding the improvement of skin structure in SD through microneedling can be confirmed taking into account the above-mentioned limitations. Further studies should also address the individual response to microneedling therapy between older and younger subjects. The specialist literature indicates that younger skin has a more pronounced regenerative capacity after microneedling therapies compared to older dermis³⁶.

In summary the prospective study demonstrates that microneedling with the EXCEED Pen is an effective minimally invasive option for improving skin texture in striae distensae. The use of standardized measurement protocols and reproducible treatment algorithms has created an objective data basis for the first time complementing previous subjective evaluation systems. Further research should investigate optimized therapy intervals and combinations with adjuvant procedures (e.g., PRP) alongside larger test groups and inclusion of an inter- or intraindividual control group where nontreated striae should be compared with treated striae distensae⁶. Demographic differences in age, gender and skin type should be minimized by using similar subject profiles to reduce variability in results.

Conclusion

The study shows that microneedling treatment with the EXCEED Medical Needling Pen can cause

significant changes in various skin parameters in cases of stretch marks (striae). In particular it was found that some parameters such as the anisotropy index, skin smoothness (SEsm) and wrinkle depth (SEw) improve over the course of treatment and approach the values of intact skin. This suggests that the therapy is effective in improving skin structure and quality in striae. However the results vary greatly between patients and not all parameters show a significant change. In addition the effect is time-dependent with some improvements only becoming visible after several weeks. Limiting factors include the small sample size and the absence of an intraindividual control group comparing non-treated striae with treated striae. No other therapies or personal care products were used in this study. Overall the study supports the effectiveness of the microneedling pen as a possible treatment option for improving stretch marks.

Conflict of Interest: The needling pens used in this study were provided free of charge by amia med (MT.DERM GmbH). Apart from this material support the authors have no financial or non-financial competing interests to disclose in relation to this manuscript.

Ethics Committee: Bavarian State Medical Association, Mühlbauerstraße 16, 81677 Munich (Ethics application number: 2023-1117).

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